

Lecture 3

Last time: smooth unconstrained optimization

Today: smooth optimization with linear equality constraints

$$\begin{array}{ll} \min_x & f(x) \quad f \in \mathcal{C}^2 \\ \text{subj. to:} & Ax = b \quad A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m \end{array}$$

(so we have m equality constraints)

How can we interpret Ax ?

$$A = \begin{pmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{pmatrix} \quad \text{where } a_i \in \mathbb{R}^m \text{ are columns of } A$$

then we see that

$$Ax = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \sum_{i=1}^n x_i a_i$$

$$\left(\begin{array}{cc} a_1 & a_n \\ 1 & 1 \end{array} \right) \left(\begin{array}{c} \vdots \\ x_n \end{array} \right) \quad i=1$$

→ Define range and null space:

$$\text{range}(A) = \{ Ax \mid x \in \mathbb{R}^n \} \subseteq \mathbb{R}^m$$

When is $\text{range}(A) = \mathbb{R}^m$? → when A is full rank

$$\rightarrow \text{rank } A = \dim \text{range}(A) = \min(m, n)$$

$$\text{null space: } \mathcal{N}(A) = \{ x \mid Ax = 0 \text{ and } x \in \mathbb{R}^n \}$$

⇒ Rank-nullity theorem:

$$\text{rank}(A) + \dim \mathcal{N}(A) = n$$

Why does this matter? → today we'll consider the three possible cases of full rank A

e.g., suppose A is wide ($m < n$) and full rank,

$$\text{then } \dim \mathcal{N}(A) = n - m$$

$$\text{thus, } \exists x_2 \in \mathcal{N}(A) \text{ s.t. } Ax_2 = 0$$

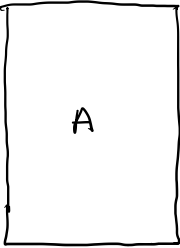
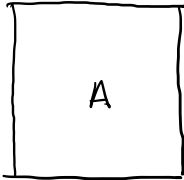
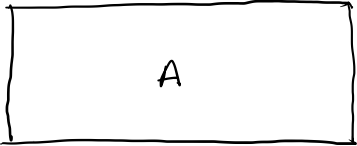
$$\text{then if we have some } x_1 \text{ s.t. } Ax_1 = b$$

$$\text{note that } A(x_1 + x_2) = Ax_1 + \cancel{Ax_2}^0 = Ax_1 = b$$

→ on this example, there are many solutions to $Ax = b$

→ on this example, there are many solutions to $Ax=b$

Three cases: only consider full rank A

<u>Tall</u>	<u>Square</u>	<u>Wide</u>
		
$m > n$	$m = n$	$m < n$
$\text{rank } A = n$	$\text{rank } A = n$	$\text{rank } A = m$
$\dim \mathcal{N}(A) = 0$	$\dim \mathcal{N}(A) = 0$	$\dim \mathcal{N}(A) = n - m$

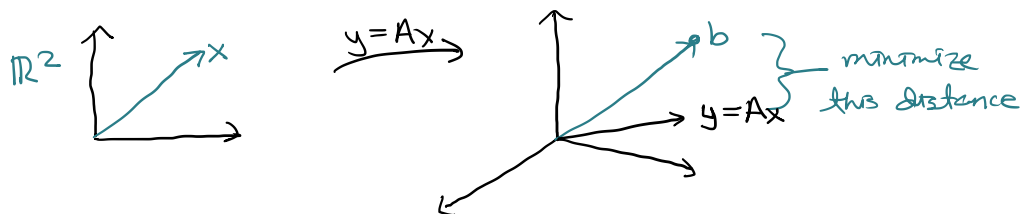
Case 1: A is square and full rank

$$\min_x f(x) \rightarrow x^* = A^{-1}b$$

subj. to: $Ax=b$

Case 2: A is tall and full rank

Consider $A \in \mathbb{R}^{3 \times 2}$



→ Cannot guarantee that $\exists x$ such that $Ax=b$

→ will instead find closest point to b

Linear least squares: minimize square of residual $r(x) = Ax - b$

$$\rightarrow \min_x \frac{1}{2} \|r(x)\|_2^2 = \min_x \frac{1}{2} \|Ax - b\|_2^2$$

$$\min_x \frac{1}{2} \|r(x)\|_2^2 = \min_x \frac{1}{2} \|Ax - b\|_2^2$$

$$\Rightarrow \frac{1}{2} \|Ax - b\|_2^2 = \frac{1}{2} (Ax - b)^T (Ax - b)$$

$$= \frac{1}{2} (x^T A^T A x - \underbrace{b^T A x}_{\text{these terms are the same}} - \underbrace{x^T A^T b}_{\text{these terms are the same}} + b^T b)$$

these terms are the same

$$(b^T A x)^T = x^T A^T b \quad \text{since these are scalars}$$

$$= \frac{1}{2} (x^T A^T A x - 2(A^T b)^T x + b^T b)$$

Now apply necessary conditions of optimality:

$$\nabla_x \left(\frac{1}{2} \|Ax - b\|_2^2 \right) := 0$$

$$= \nabla_x \left(\frac{1}{2} (\underline{x^T A^T A x} - 2 \underline{(A^T b)^T x} + \underline{b^T b}) \right)$$

$$= \frac{1}{2} (2 A^T A x^* - 2 A^T b + 0)$$

$$= A^T A x^* - 2 A^T b$$

$$\Rightarrow x^* = (A^T A)^{-1} A^T b \quad \text{Linear least squares}$$

Nonlinear least squares: instead of a linear equality

$Ax = b$, what if we had a nonlinear expression

$$l(x) = b? \rightarrow \min_x \frac{1}{2} \|l(x) - b\|_2^2$$

Linear least squares has many applications in robotics, controls, signal processing, etc.

Nonlinear least squares (NLLS) is commonly used within SLAM